

Resin Composite Restorative Materials (Practical)

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Manipulation of Resin Composite

1. Pulpal protection:

- ▶ In very deep cavities, protect the pulp by using liner.

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2. Etching and bonding:

1. Bonding to Enamel:

a) Acid etching:

By applying 37% phosphoric acid for 15 to 30 seconds.

Importance of acid etching:

1. Increase surface energy of enamel.
2. Cleaning enamel surface from debris.
3. Increasing surface area of the exposed enamel to resin.
4. Forming micro-pores that are penetrated by adhesive to form resin tags.

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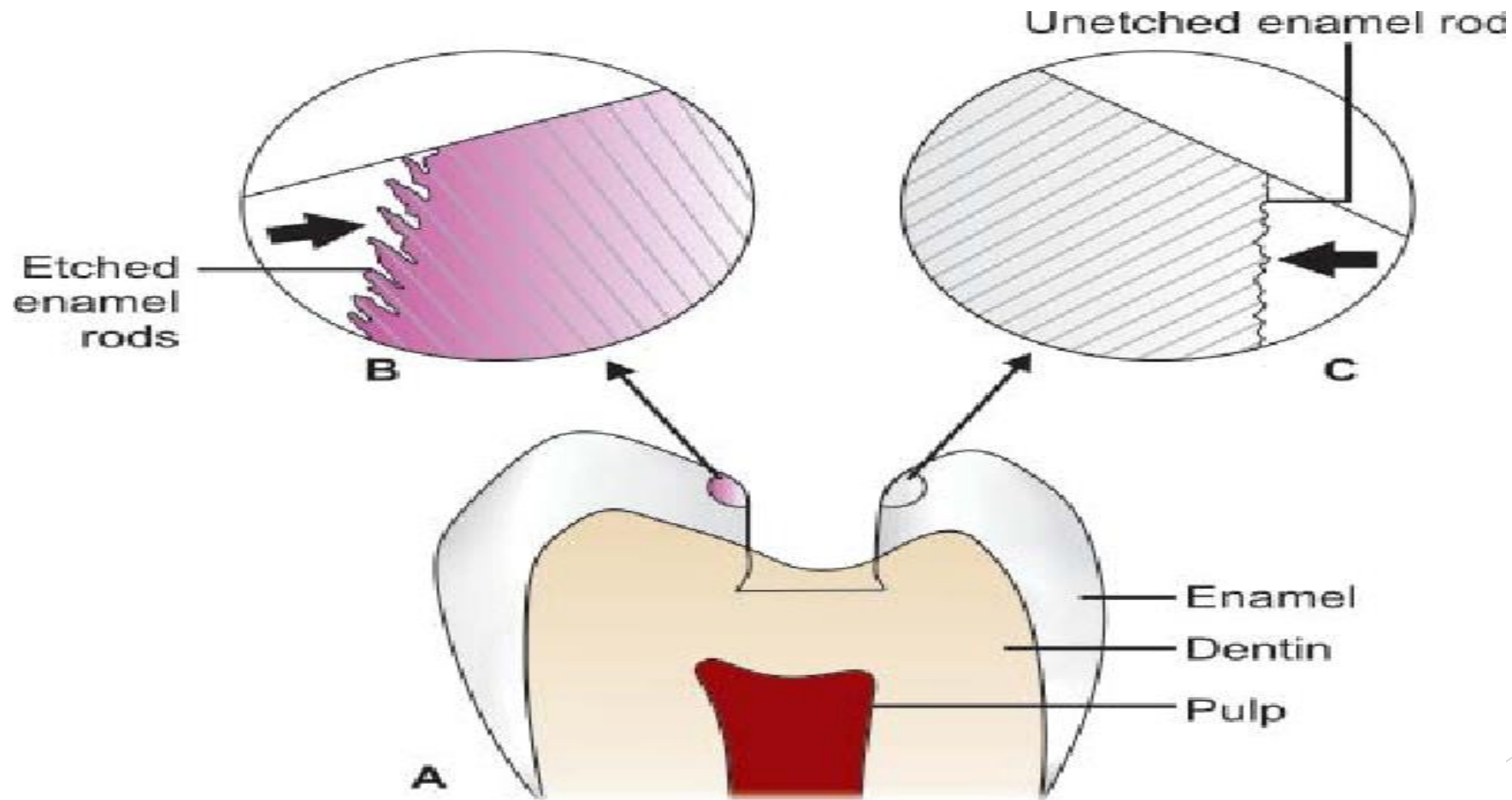


Fig. 10.11. Diagram illustrating the manipulation of resin composite on enamel and dentin.

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1. Bonding to Enamel:

b) Adhesive application:

An adhesive (bonding agent)is applied to fulfill the micropores.

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2. Bonding to dentin:

- a) **Acid conditioning** (mild acid for less time than enamel):
 - ▶ Remove smear layer.
 - ▶ Demineralize dentin causes collapse of dentinal tubules and decreasing surface energy.

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2. Bonding to dentin:

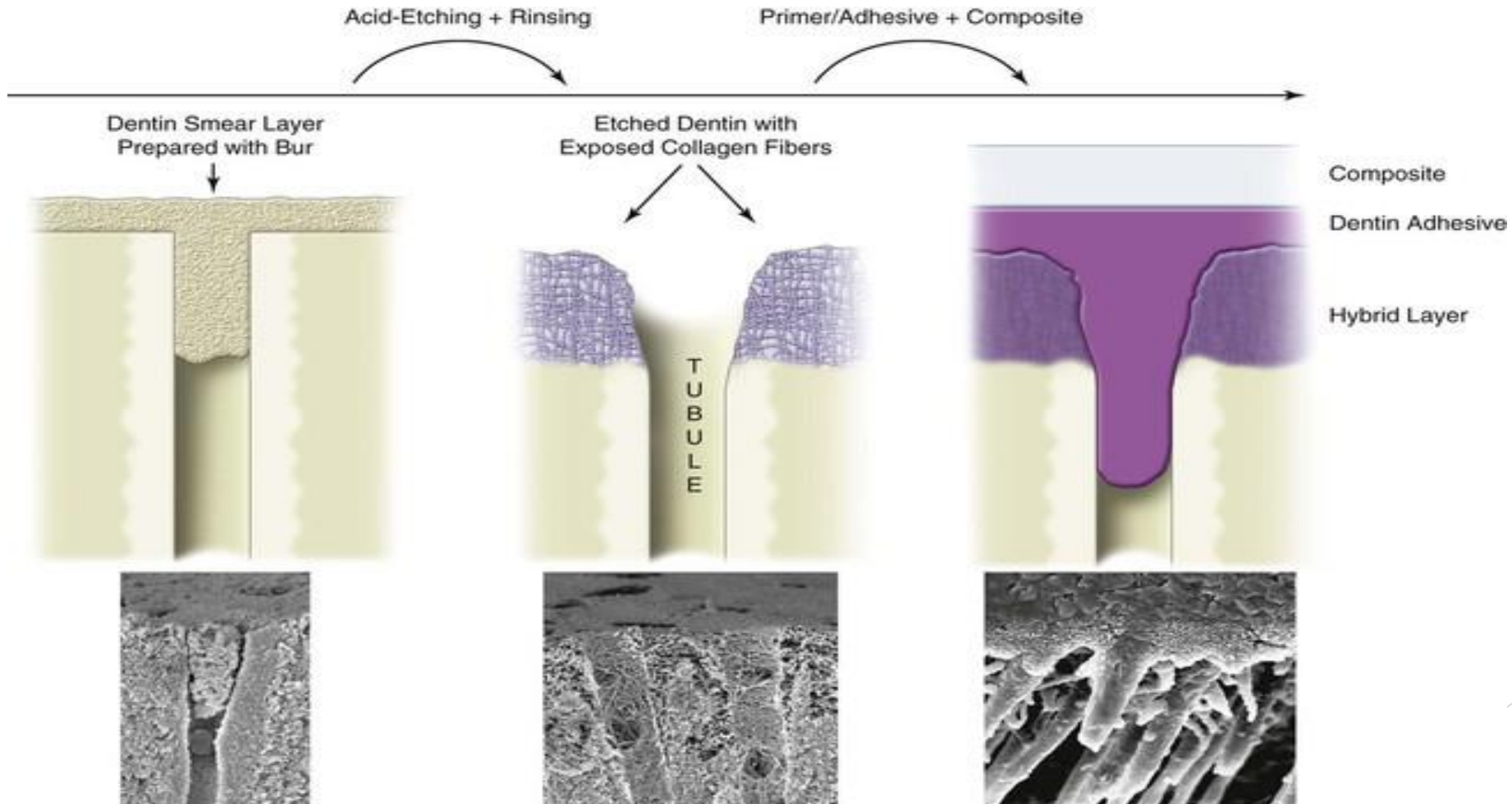
b) Primer:

It is a molecule with two functional groups:

1. Hydrophilic group: interact with moist dentine ,reopen collapsed dentinal tubules so increasing surface energy.
2. Hydrophobic group: interact with adhesive.

c) Adhesive (Bonding agent):

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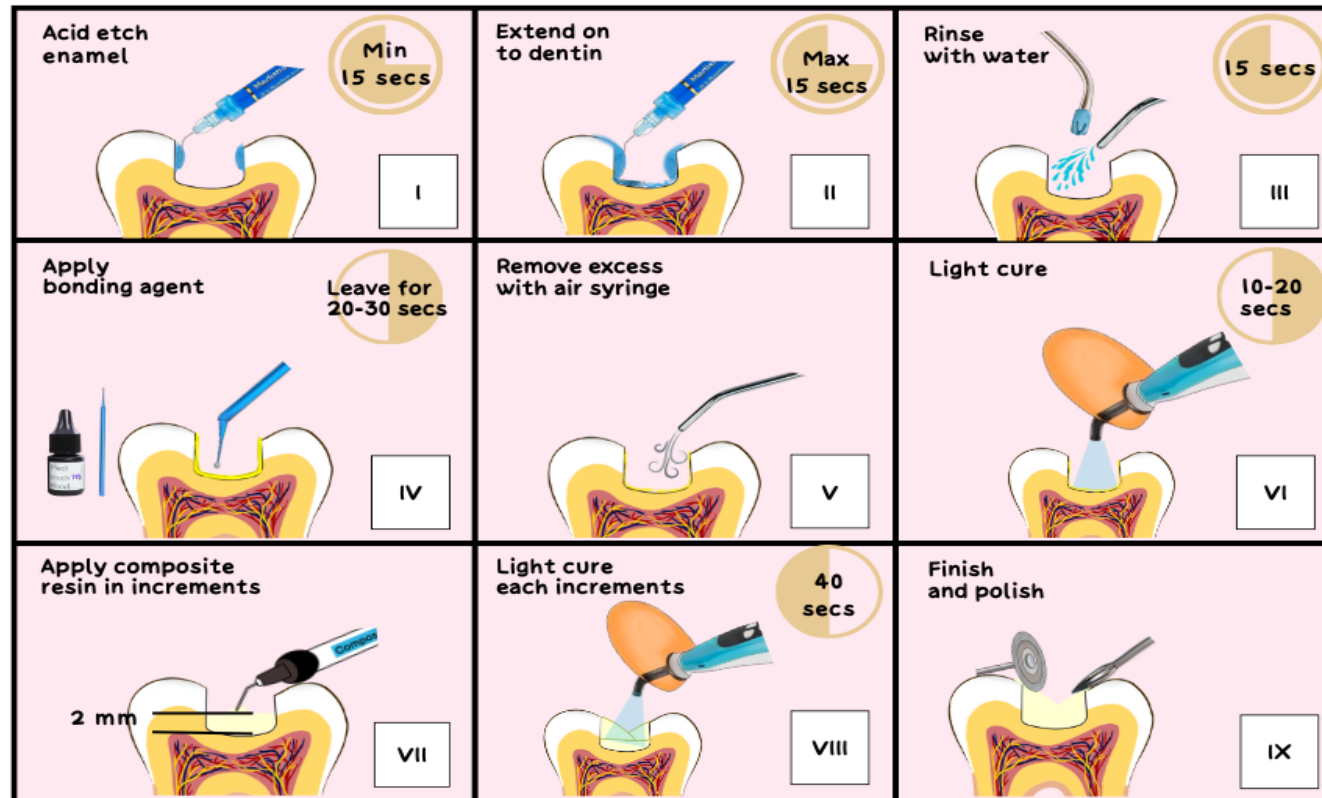
2. Bonding to dentin:

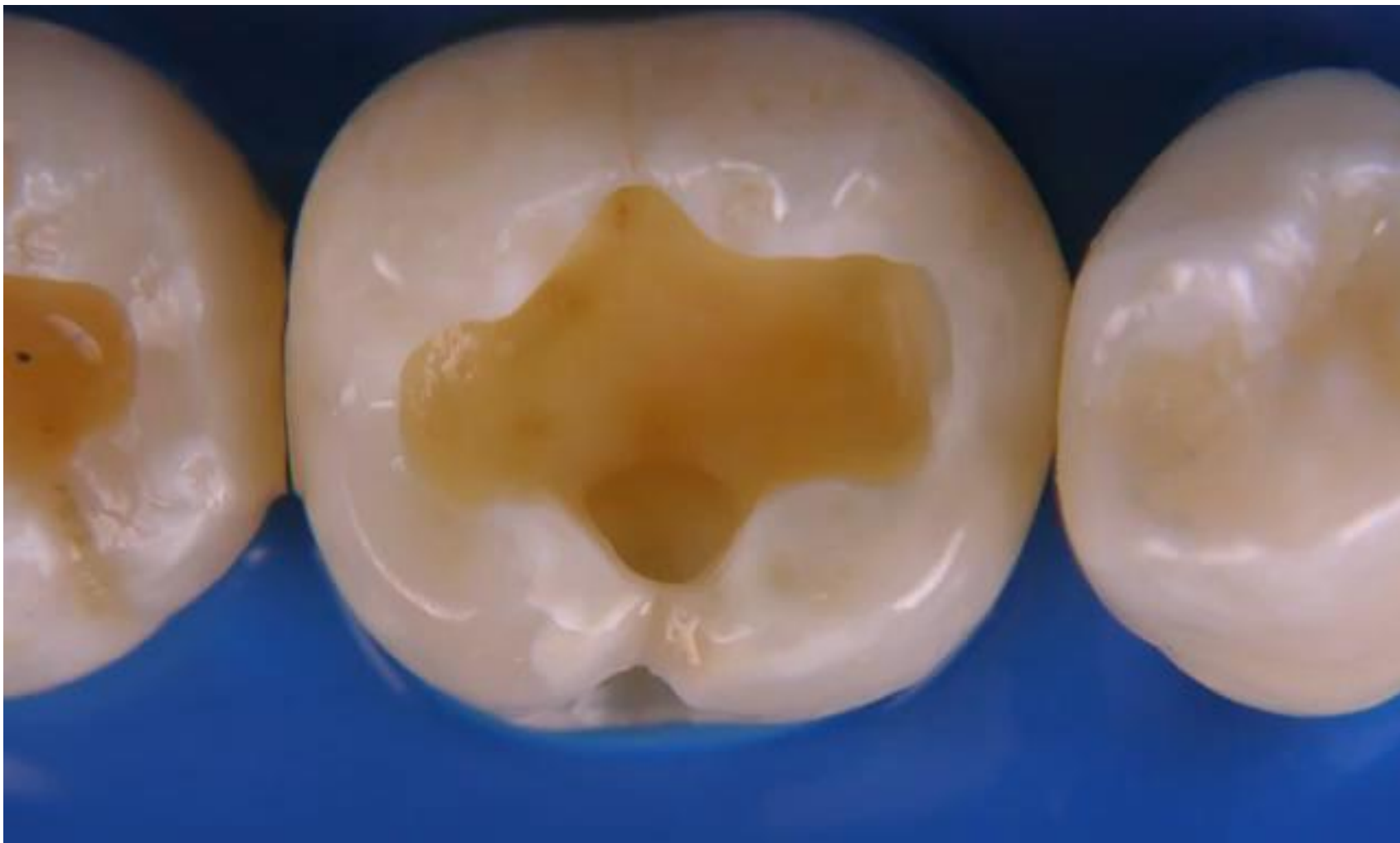
- ▶ Newer generations of adhesive systems had been evolved with different techniques.

3-step Etch-and-rinse	 Etchant	 Primer	 Adhesive
2-step Etch-and-rinse	 Etchant	 Single-bottle primer/adhesive	
2-step Self-etching	 Self-etching primer		 Adhesive
1-step Self-etching	 Self-etching primer/adhesive		

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Steps in application of composite





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3. Dispensing:

Light cured resin composite:

It is supplied as one paste system.

The form of supply :

- ▶ Screw-type syringe.
- ▶ Compule (single dose) with special gun.
- ▶ Syringe with dispensing tip for flowable composite.

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3. Dispensing:



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3. Dispensing:

Chemical cured resin composite:

It is supplied in two pastes.

The form of supply:

- ▶ Two jars with two different colors to insure homogenous mixing.

The two pastes are mixed over paper pad using plastic, Teflon or gold coated spatula.

- ▶ Gun, cartridge of two tubes and mixing tip system

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3. Dispensing:



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4. Placement:

- ▶ Resin composite placed inside the cavity using plastic , Teflon or gold coated instrument.
- ▶ Compules and flowable resin composites are placed directly inside the prepared cavity.



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5. Polymerization:

a) Light cured composite:

- ▶ Restoration should be sufficiently cured to avoid the unpolymerized material in the base of the cavity.
- ▶ The unreacted monomers irritate the pulp and reduced the mechanical properties of the restoration.

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5. Polymerization:

a) Light cured composite:

- ▶ Depth of curing is affected by :



1. Type of curing unit

- ▶ As they differ in light intensity and wavelength.

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5. Polymerization:

a) Light cured composite:

- Depth of curing is affected by :

2. Resin composite shade:

Darker shades have shallower light penetration.

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5. Polymerization:

a) Light cured composite:

- Depth of curing is affected by :

3. Distance between the light source and restoration:

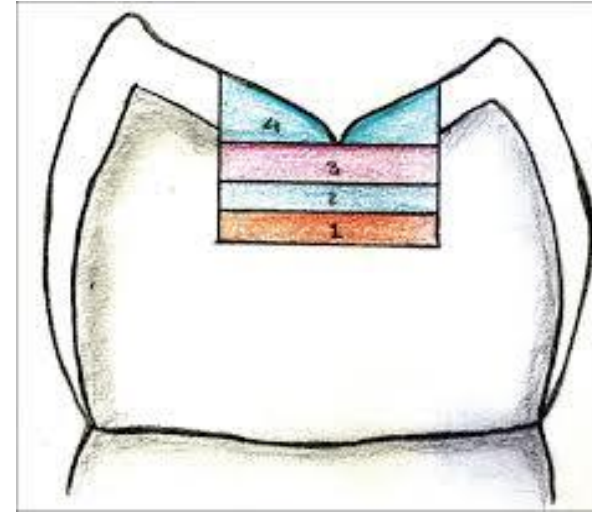
it should be close as much as possible to the restoration.

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5. Polymerization:

a) **Light cured composite:**

- Depth of curing is affected by :



4. Resin composite thickness:

The thickness of each increment should be 2 mm.
(to decrease polymerization shrinkage).

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5. Polymerization:

a) Light cured composite:

- Depth of curing is affected by :

5. Exposure time:

Increasing exposure time will increase curing of the restoration.

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5. Polymerization:

a) Light cured composite:

- Depth of curing is affected by :

6. Initiator system :

The wavelength of the curing unit should be compatible with the initiator of the restoration.

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5. Polymerization:

b) chemical cured composite:

- It can be placed in one increment.
- Working and setting time about 4-5 minutes.

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5. Polymerization:

c) Dual cured composite:

- ▶ Polymerization initiated by light and continued chemically.
- ▶ Recommended when part of resin composite is not accessible to the curing light.

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6.Finishing and Polishing:

- ▶ The outer layer of the resin composite is uncured due to the presence of the oxygen and called air inhibited layer.
- ▶ This layer should be removed due to its lower mechanical properties and high affinity to staining.

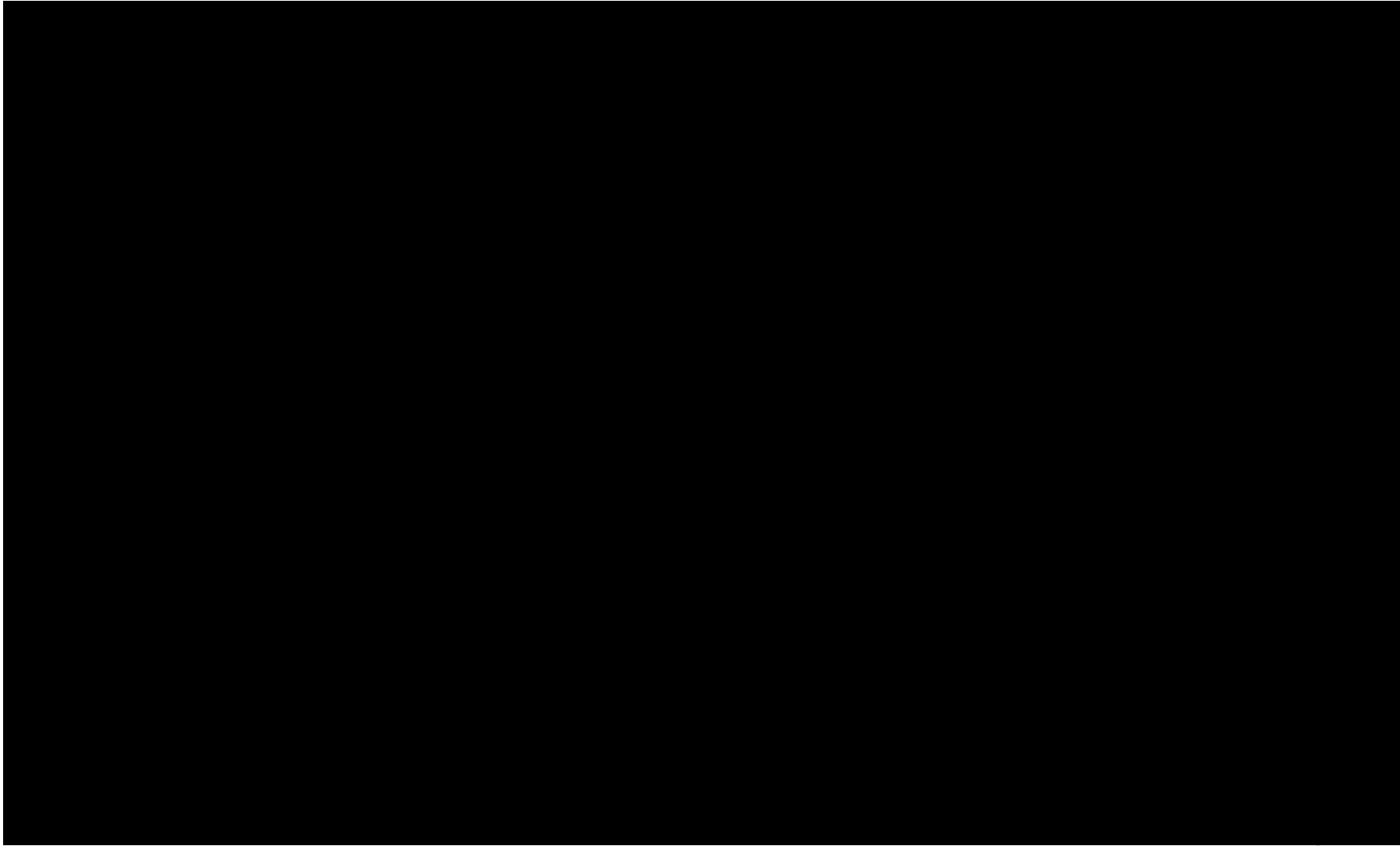
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6. Finishing and Polishing:

- ▶ Finishing and polishing is done by finishing burs.
- ▶ Finishing and polishing starts with coarse tool then gradually progress to the fine tool.
- ▶ Avoid overheating the restoration during finishing and polishing.

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Thank You